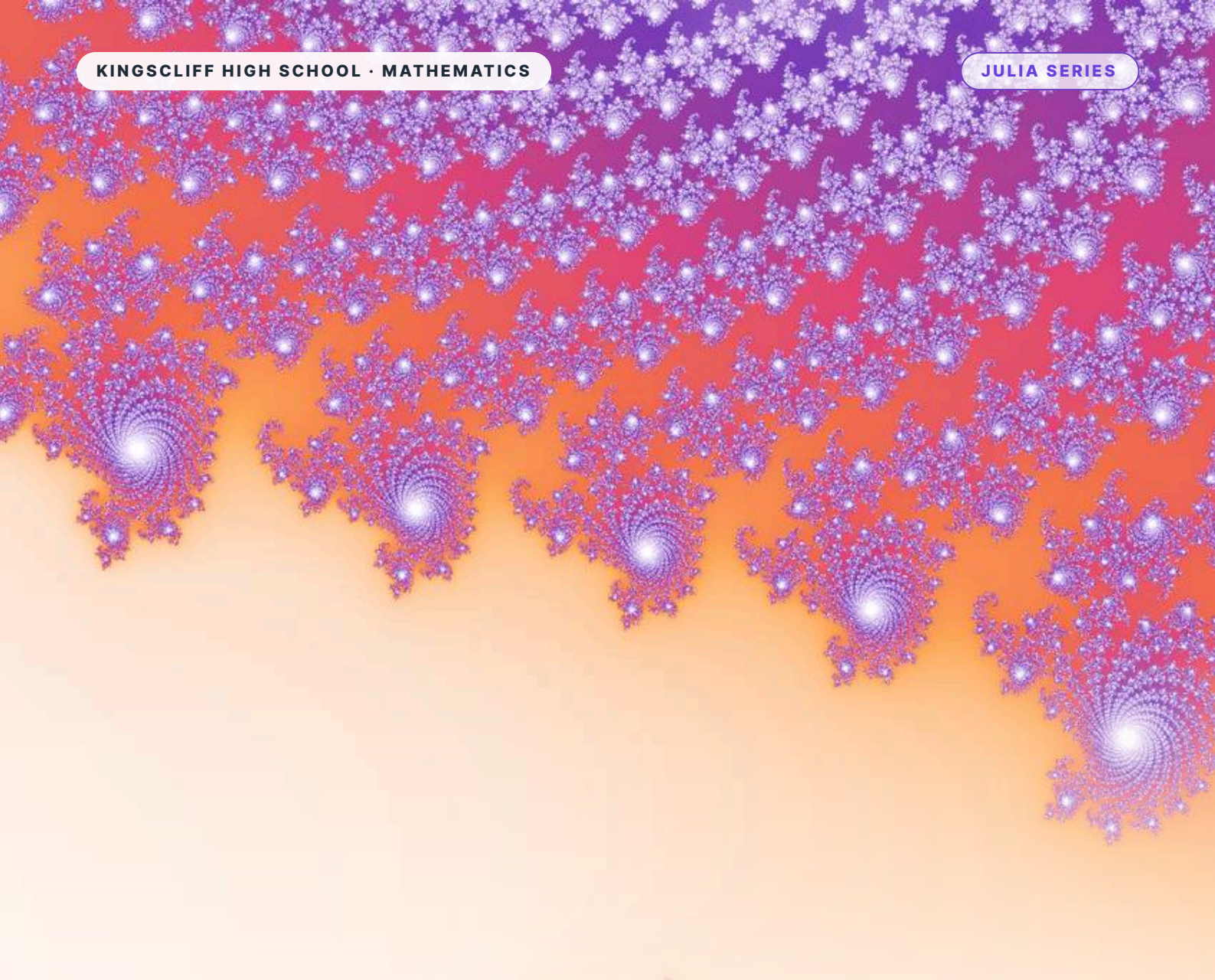


KINGSCLIFF HIGH SCHOOL · MATHEMATICS

JULIA SERIES

YEAR 10 STANDARD → YEAR 11 MATHEMATICS STANDARD

Summer Maths

Ready for Year 11

Six weeks. Twenty questions a week. Every Year 10 skill that Year 11 Mathematics Standard builds on – mixed together so it sticks.

Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6

NAME _____ CLASS _____

How it works

One page a week

Each week has **20 questions** and a **bonus**. Set aside about **40 minutes** – one sitting or a few short ones. Questions start easier and get a little harder through the page.

The same pattern every week

Question 1 is always number and rounding, question 11 is always Pythagoras, and so on. If the same number keeps going wrong, that is the skill to revise before Year 11.

1	Number & rounding	11	Pythagoras
2	Percentages	12	Trigonometry
3	Rates & ratios	13	Area & surface area
4	Expand & factorise	14	Volume & capacity
5	Substitute into formulas	15	Units & error
6	Change the subject	16	Earning money
7	Equations	17	Interest & depreciation
8	Equations from words	18	Data
9	Gradient & lines	19	Probability
10	Linear models	20	Problem solving

Why Julia? Each banner is part of the same *Julia set*: pick a number c, then repeat $z \to z^2 + c$ for every point. Points that stay small are coloured in. Every booklet in the series zooms into a different corner of the same set – this one looks along the valley between two spirals.

NSW syllabus (NESA Mathematics K–10, 2022). Revises MA5-FIN-C-01, MA5-FIN-C-02, MA5-ALG-C-01, MA5-ALG-P-01, MA5-EQU-C-01, MA5-EQU-P-02, MA5-LIN-C-01, MA5-LIN-C-02, MA5-RAT-P-01, MA5-MAG-C-01, MA5-TRG-C-01, MA5-TRG-C-02, MA5-TRG-P-01, MA5-ARE-C-01, MA5-ARE-P-01, MA5-VOL-C-01, MA5-VOL-P-01, MA5-DAT-C-01, MA5-DAT-C-02, MA5-PRO-C-01, with MA4-PYT-C-01. ★ Year 11 preview: Mathematics Standard 11–12 (2024) focus areas Time and location; Networks, paths and trees; Earning money; Managing money.

Rules of the game

- **No calculator.** Leave answers exact (e.g. 24π) unless asked to round. Trig questions give you any value you need.
- **Show working** in the space under each question.
- **Mark it yourself** using the answers at the back (pages 10–12). Each answer has a line of working to help you find a mistake.
- **Fix it.** Redo anything you got wrong a day or two later, without looking.
- ★ questions preview Year 11 Mathematics Standard (Stage 6). Have a go – they are the ones that will help most in Term 1.
- **Stuck?** The hints page (page 9) has the key facts and formulas.

My summer tracker

WEEK	DATE	SCORE	DONE	BONUS	REVISE (Q NUMBERS)
1 Week 1		/ 20	☐	☐	
2 Week 2		/ 20	☐	☐	
3 Week 3		/ 20	☐	☐	
4 Week 4		/ 20	☐	☐	
5 Week 5		/ 20	☐	☐	
6 Week 6		/ 20	☐	☐	

First up in Year 11 Mathematics Standard: formulas and equations, linear relationships, earning and managing money, applications of measurement, time and location, networks, and data analysis. Every one of them starts from the skills in these six pages.

WEEK 1

Summer Maths

Ready for Year 11

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Algebra · Equations · Linear · Measurement · Money · Data & chance · In context

★ Year 11 preview (Stage 6 Standard) · No calculator

1 NUMBER & ROUNDING

Round 3.4786 to 2 decimal places.

2 PERCENTAGES

Find 12.5% of \$640.

3 RATES & RATIOS

Simplify the ratio $45 : 60$

4 EXPAND & FACTORISE

Expand and simplify $4(x - 3) - 2(x - 5)$

5 SUBSTITUTE INTO FORMULAS

$v = u + at$. Find v when $u = 5$, $a = -2$ and $t = 3$.

6 CHANGE THE SUBJECT

Make x the subject of $y = 3x - 6$.

7 EQUATIONS

Solve $6x - 7 = 2x + 9$

8 EQUATIONS FROM WORDS

Three times a number, decreased by 4, is 17. Find the number.

9 GRADIENT & LINES

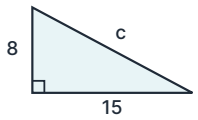
State the gradient and y-intercept of $y = 5 - 2x$.

10 LINEAR MODELS

A plumber charges \$90 call-out plus \$70 an hour. Write a formula for the cost C of an h-hour job, and find the cost of a 3-hour job.

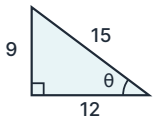
11 PYTHAGORAS

Find c. (cm)



12 TRIGONOMETRY

Write $\cos\theta$ as a fraction in simplest form.



13 AREA & SURFACE AREA

Find the area of a trapezium with parallel sides 8 m and 12 m and perpendicular height 5 m.

14 VOLUME & CAPACITY

A tank is 3 m by 2 m by 1.5 m. Find its capacity in kilolitres. ($1\ \text{m}^3 = 1$ kL)

15 UNITS & ERROR

Convert 3.6 m² to cm².

16 EARNING MONEY

Mai earns \$26.50 an hour. Find her pay for 20 hours.

17 INTEREST & DEPRECIATION

Find the simple interest on \$3000 at 4% p.a. for 5 years.

18 DATA

Find the mean of $23,\ 27,\ 31,\ 25,\ 29,\ 33$.

19 PROBABILITY

A card is drawn at random from cards numbered 1 to 20. Find $P(\text{multiple of } 5)$.

20 PROBLEM SOLVING ★

In winter, Sydney time is 2 hours ahead of Perth time. A flight leaves Perth at 11:30 am (Perth time) and takes 4 h 15 min. What is the time in Sydney when it lands?

BONUS

A number is doubled, then 5 is added, then the result is tripled. The answer is 51. Find the number.

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Algebra · Equations · Linear · Measurement · Money · Data & chance · In context

★ Year 11 preview (Stage 6 Standard) · No calculator

1 NUMBER & ROUNDING
Round $0.006\,82$ to 2 significant figures.

2 PERCENTAGES
A \$1200 TV is reduced by 15%. Find the sale price.

3 RATES & RATIOS
A car uses 24 L of fuel to travel 300 km. Find its fuel consumption in L/100 km.

4 EXPAND & FACTORISE
Expand $(x+4)(x-6)$

5 SUBSTITUTE INTO FORMULAS
Use $d = st$ to find the distance travelled at 85 km/h for 2.5 hours.

6 CHANGE THE SUBJECT
Make t the subject of $v = u + at$.

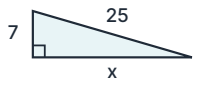
7 EQUATIONS
Solve $\dfrac{x-4}{3} = 5$

8 EQUATIONS FROM WORDS
Tom is 5 years older than Ana. Their ages add to 39. How old is Ana?

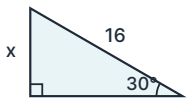
9 GRADIENT & LINES
Find the gradient of the line through $(1,\ 3)$ and $(5,\ 11)$.

10 LINEAR MODELS
A \$900 phone loses \$150 of value each year. Write a rule for its value V after n years. When is it worth nothing?

11 PYTHAGORAS
Find x. (cm)



12 TRIGONOMETRY
Use $\sin 30^\circ = 0.5$ to find x. (cm)



13 AREA & SURFACE AREA
Find the area of a circle with diameter 12 cm, in terms of π.

14 VOLUME & CAPACITY
Find the volume of a cylinder with radius 5 cm and height 8 cm, in terms of π.

15 UNITS & ERROR
A length is measured as 24 cm, to the nearest centimetre. What is the absolute error?

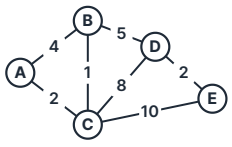
16 EARNING MONEY
Raj's normal rate is \$32 an hour. Find his pay for 38 normal hours and 5 hours at time-and-a-half.

17 INTEREST & DEPRECIATION
\$5000 is invested at 6% p.a. compound interest. Find its value after 2 years.

18 DATA
Find the median of $14,\ 9,\ 21,\ 17,\ 12,\ 16,\ 19,\ 11$.

19 PROBABILITY
Two dice are rolled. Find $P(\text{the sum is } 10)$.

20 PROBLEM SOLVING ★
The numbers show travel times in minutes. Find the shortest path from A to E and its length.



Not to scale

BONUS A rectangle has a perimeter of 34 cm and a diagonal of 13 cm. Find its length and width.

WEEK 3

Summer Maths

Ready for Year 11

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Algebra · Equations · Linear · Measurement · Money · Data & chance · In context

★ Year 11 preview (Stage 6 Standard) · No calculator

1 NUMBER & ROUNDING

Write $0.000\,032$ in scientific notation.

2 PERCENTAGES

A house bought for \$640 000 sells for \$720 000. Find the profit as a percentage of the cost.

3 RATES & RATIOS

Share \$900 in the ratio $2 : 3 : 4$.

4 EXPAND & FACTORISE

Factorise $6x^2 - 9x$

5 SUBSTITUTE INTO FORMULAS

$S = \frac{n}{2}(a + l)$. Find S when $n = 20$, $a = 3$ and $l = 41$.

6 CHANGE THE SUBJECT

Make r the subject of $A = \pi r^2$ (where $r > 0$).

7 EQUATIONS

Solve $3(2x + 1) = 4(x + 5)$

8 EQUATIONS FROM WORDS

Adult tickets cost \$25 and child tickets \$15. 40 tickets were sold for \$800. How many were adult tickets?

9 GRADIENT & LINES

Find the equation of the line with gradient 3 that passes through $(2,\ 1)$.

10 LINEAR MODELS

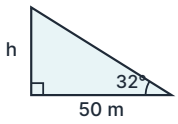
A café's weekly costs are $C = 200 + 4n$ and its income is $R = 9n$, where n is the number of coffees. How many coffees must it sell to break even?

11 PYTHAGORAS

A ramp rises 1.2 m over a horizontal distance of 3.5 m. How long is the ramp?

12 TRIGONOMETRY

From 50 m away, the angle of elevation of the top of a building is $32°$. Use $\tan 32° \approx 0.62$ to find its height.



13 AREA & SURFACE AREA

Find the surface area of a closed cylinder with radius 4 cm and height 6 cm, in terms of π.

14 VOLUME & CAPACITY

Find the volume of a cone with radius 3 cm and height 8 cm, in terms of π.

15 UNITS & ERROR

A 50 m length has an absolute error of 0.5 m. Find the percentage error.

16 EARNING MONEY

Ali earns \$450 a week plus 3% commission on sales. Find his pay in a week with \$12 000 of sales.

17 INTEREST & DEPRECIATION

A \$24 000 car depreciates by 25% a year. Find its value after 2 years.

18 DATA

Find the interquartile range of $4,\ 7,\ 8,\ 10,\ 12,\ 15,\ 18,\ 20,\ 21$.

19 PROBABILITY

A bag holds 4 red and 6 blue balls. Two are drawn without replacement. Find $P(\text{both blue})$.

20 PROBLEM SOLVING ★

Melbourne is at about $145°$E and London at $0°$. Using $15°$ of longitude = 1 hour, find the time difference.

BONUS Find the whole number n for which $n^2 + (n+1)^2 = 145$.

WEEK 4

Summer Maths

Ready for Year 11

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Algebra · Equations · Linear · Measurement · Money · Data & chance · In context

★ Year 11 preview (Stage 6 Standard) · No calculator

1 NUMBER & ROUNDING

Evaluate $(4.8 \times 10^6) \div (1.2 \times 10^2)$ in scientific notation.

2 PERCENTAGES

A price including 10% GST is \$660. Find the price before GST.

3 RATES & RATIOS

Convert 90 km/h to metres per second.

4 EXPAND & FACTORISE

Expand $(2x - 3)^2$

5 SUBSTITUTE INTO FORMULAS

Use $A = P(1 + r)^n$ to find A when $P = 2000$, $r = 0.05$ and $n = 2$.

6 CHANGE THE SUBJECT

Make h the subject of $V = \frac{1}{3}\pi r^2 h$.

7 EQUATIONS

Solve $5 - 2x = 3(x - 5)$

8 EQUATIONS FROM WORDS

Two numbers add to 50 and differ by 12. Find them.

9 GRADIENT & LINES

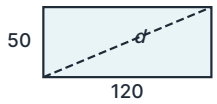
Are the lines $y = 2x + 1$ and $y = -\frac{1}{2}x + 4$ perpendicular? Explain.

10 LINEAR MODELS

A 30 cm candle burns down 2.5 cm each hour. Write a formula for its height h after t hours. When will it be 10 cm tall?

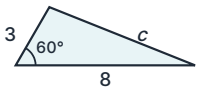
11 PYTHAGORAS

A TV screen is 120 cm wide and 50 cm high. Find the length of its diagonal.



12 TRIGONOMETRY

Use the cosine rule to find c. $(\cos 60^\circ = 0.5)$



13 AREA & SURFACE AREA

Find the area of a sector with radius 6 cm and angle 60°, in terms of π.

14 VOLUME & CAPACITY

Find the volume of a sphere with radius 3 cm, in terms of π.

15 UNITS & ERROR

A length is 7.4 cm, measured to the nearest millimetre. Between what two values does the true length lie?

16 EARNING MONEY

Raj earns \$68 000 a year and has \$2500 of allowable tax deductions. Find his taxable income.

17 INTEREST & DEPRECIATION

On \$10 000 for 2 years, which earns more interest: 10% p.a. simple or 9% p.a. compound?

18 DATA

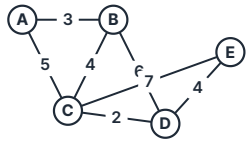
For $3,\ 5,\ 5,\ 6,\ 8,\ 40$, which measure of centre best describes the data? Explain.

19 PROBABILITY

Events A and B are independent, with $P(A) = 0.6$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

20 PROBLEM SOLVING ★

The numbers are cable lengths in km. Find the length of the minimum spanning tree that connects every town.



Not to scale

BONUS A rectangle is $(x + 5)$ cm long and $(x - 2)$ cm wide. Its area is 30 cm². Find x.

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Algebra · Equations · Linear · Measurement · Money · Data & chance · In context

★ Year 11 preview (Stage 6 Standard) · No calculator

1 NUMBER & ROUNDING

Round $26\,849$ to 3 significant figures.

2 PERCENTAGES

A salary of \$70 000 rises by 3%. Find the new salary.

3 RATES & RATIOS

A 1.25 L drink costs \$2.50 and a 2 L drink costs \$3.60. Which is the better buy?

4 EXPAND & FACTORISE

Expand and simplify $(x+5)(x-5)-x(x-2)$

5 SUBSTITUTE INTO FORMULAS

Use $F = 1.8C + 32$ to find C when $F = 50$.

6 CHANGE THE SUBJECT

Make b the subject of $P = 2(a+b)$.

7 EQUATIONS

Solve $2(3x-1)-(x+4)=9$

8 EQUATIONS FROM WORDS

A rectangle is 4 m longer than it is wide, and its perimeter is 48 m. Find its area.

9 GRADIENT & LINES

Find the x- and y-intercepts of $3x+4y=12$.

10 LINEAR MODELS

Use 1 AUD = 0.65 USD. How many US dollars is 400 AUD? How many Australian dollars is 130 USD?

11 PYTHAGORAS

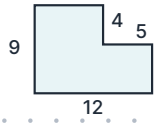
A box is 6 cm by 8 cm by 24 cm. Find the diagonal of its 6 × 8 base, then the longest rod that fits in the box.

12 TRIGONOMETRY

In a right-angled triangle, the adjacent side is 12 and the hypotenuse is 13. Write $\cos\theta$ and find $\tan\theta$.

13 AREA & SURFACE AREA

Find the area. (m)



14 VOLUME & CAPACITY

A pool is 25 m long, 10 m wide and 1.2 m deep. How many kilolitres does it hold?

15 UNITS & ERROR

A 2 kg bag is weighed to the nearest 10 g. Find the percentage error.

16 EARNING MONEY

Sophie earns \$1500 a week. She takes 4 weeks' leave with 17.5% leave loading. How much is the loading?

17 INTEREST & DEPRECIATION

A \$9000 machine loses \$1200 of value each year (straight-line depreciation). Find its value after 4 years.

18 DATA

A scatterplot of hours studied against test mark has points rising from left to right, close to a straight line. Describe the association.

19 PROBABILITY

A coin is tossed 200 times and lands heads 92 times. Find the relative frequency of heads.

20 PROBLEM SOLVING ★

Ava earns \$860 a week. She spends 30% on rent, \$160 on food and \$120 on transport. How much is left?

BONUS

A \$20 000 car loses 20% of its value each year. After how many whole years is it first worth less than \$10 000?

WEEK 6

Summer Maths

Ready for Year 11

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Algebra · Equations · Linear · Measurement · Money · Data & chance · In context

★ Year 11 preview (Stage 6 Standard) · No calculator

1 NUMBER & ROUNDING

Evaluate $(2.4 \times 10^{-3}) \times (5 \times 10^{5})$ in scientific notation.

2 PERCENTAGES

After a 25% increase, a price is \$150. Find the original price.

3 RATES & RATIOS

A tap fills a 2.4 kL tank at 15 L per minute. How long does it take? Answer in hours and minutes.

4 EXPAND & FACTORISE

Factorise $x^2 - x - 20$

5 SUBSTITUTE INTO FORMULAS

$A = \frac{h}{2}(a + b)$. Find a when $A = 60$, $h = 6$ and $b = 12$.

6 CHANGE THE SUBJECT

Make r the subject of $V = \frac{4}{3}\pi r^3$.

7 EQUATIONS

Solve simultaneously: $4x + 3y = 18$ and $y = 2x - 4$

8 EQUATIONS FROM WORDS

Pens cost \$2 and notebooks \$5. Mia buys 12 items for \$39. How many notebooks did she buy?

9 GRADIENT & LINES

Find the equation of the line through $(1,\ 4)$ that is perpendicular to $y = 2x + 3$.

10 LINEAR MODELS

Electricity costs \$1.10 a day plus 30 cents per kWh. Find the bill for 90 days in which 1200 kWh were used.

11 PYTHAGORAS

A cone has radius 5 cm and perpendicular height 12 cm. Find its slant height.

12 TRIGONOMETRY

In $\triangle ABC$, $a = 10$, $A = 30^\circ$ and $B = 50^\circ$. Use $\sin 30^\circ = 0.5$ and $\sin 50^\circ \approx 0.77$ to find b.

13 AREA & SURFACE AREA

A square pyramid has a base of side 10 cm, and each triangular face has a perpendicular height of 13 cm. Find its surface area.

14 VOLUME & CAPACITY

Find the volume of a hemispherical bowl with radius 6 cm, in terms of π.

15 UNITS & ERROR

A field is 120 m by 80 m, each measured to the nearest metre. Find the largest possible area.

16 EARNING MONEY ★

An employer pays 12% superannuation on a \$75 000 salary. How much super is paid in the year?

17 INTEREST & DEPRECIATION

\$8000 is invested at 5% p.a. compound interest for 3 years. Find the interest earned.

18 DATA

The mean of 8 scores is 15. A score of 29 is removed. Find the new mean.

19 PROBABILITY

A die is rolled three times. Find $P(\text{no sixes})$.

20 PROBLEM SOLVING ★

It is 9:00 pm Friday in Sydney (UTC+10). What time and day is it in Los Angeles (UTC−7)?

BONUS

\$10 000 is invested for 3 years at 10% p.a. compounded yearly. How much more is this than 10% p.a. simple interest?

Hints & formulas

Number

Sig. figs start at the first non-zero digit

Increase by 3%: $\times 1.03$ · price before GST $=$ price $\div 1.1$

Best buy: compare the cost per unit

km/h to m/s: ÷ 3.6 · fuel use (L/100 km) $= \frac{\text{litres}}{\text{km}} \times 100$

Algebra

Expand: multiply every term inside by the term outside

$(a-b)^2 = a^2 - 2ab + b^2$

Factorise: HCF first, then $x^2 + bx + c = (x+p)(x+q)$

Change the subject: undo operations in reverse order

Equations

Expand, collect like terms, then solve

Simultaneous: substitute one equation into the other

Worded problems: let the unknown be a letter; write an equation; check

Linear

$m = \frac{\text{rise}}{\text{run}}$ · $y = mx + c$

Parallel: equal gradients · perpendicular: $m_1 m_2 = -1$

Intercepts: put $x = 0$ (for y) and $y = 0$ (for x)

Linear model: starting value + rate × time

Measurement

$c^2 = a^2 + b^2$ · $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

$\frac{a}{\sin A} = \frac{b}{\sin B}$ · $c^2 = a^2 + b^2 - 2ab\cos C$

Sector $A = \frac{\theta}{360}\pi r^2$ · cylinder $SA = 2\pi r^2 + 2\pi rh$

Cone $V = \frac{1}{3}\pi r^2 h$ · sphere $V = \frac{4}{3}\pi r^3$

Absolute error $= \frac{1}{2}$ precision · % error $= \frac{\text{abs. error}}{\text{measurement}} \times 100\%$

Money

Overtime: time-and-a-half $= 1.5\times$ rate

Leave loading: 17.5% of holiday pay

Simple interest $I = Prn$ · compound $A = P(1+r)^n$

Declining balance $S = V_0(1-r)^n$ · straight line $S = V_0 - Dn$

★ Superannuation: a percentage of salary paid by the employer

Data & chance

IQR $= Q_3 - Q_1$ · outliers affect the mean most

Relative frequency $= \frac{\text{frequency}}{\text{total}}$

Independent: $P(A \text{ and } B) = P(A)P(B)$

Scatterplots: describe direction, strength and form

In context

★ 15° of longitude = 1 hour of time

★ Time zones: east is ahead (UTC+10 is 17 h ahead of UTC−7)

★ Shortest path: try every route and keep the best

★ Spanning tree: add the shortest edges, never making a loop

Answers

1 Week 1 answers

1. 3.48 third decimal is 8, so round up
2. **\$80** $12.5\% = \frac{1}{8}$
3. $3 : 4$ divide both parts by 15
4. $2x - 2$ $4x - 12 - 2x + 10$
5. $v = -1$ $5 + (-6)$
6. $x = \dfrac{y + 6}{3}$ add 6, then divide by 3
7. $x = 4$ $4x = 16$
8. **7** $3n - 4 = 17$
9. $m = -2, c = 5$ rewrite as $y = -2x + 5$
10. $C = 90 + 70h$; **\$300** $90 + 210$
11. **17 cm** $c^2 = 64 + 225 = 289$
12. $\cos\theta = \frac{4}{5}$ $\frac{\text{adjacent}}{\text{hypotenuse}} = \frac{12}{15}$
13. **50 m²** $\frac{5}{2}(8 + 12)$
14. **9 kL** $3 \times 2 \times 1.5 = 9\ \text{m}^3$
15. **36 000 cm²** $1\ \text{m}^2 = 10\,000\ \text{cm}^2$
16. **\$530** 20×26.50
17. **\$600** $I = 3000 \times 0.04 \times 5$
18. 28 $168 \div 6$
19. $\frac{1}{5}$ 5, 10, 15, 20: $\frac{4}{20}$
20. **5:45 pm** lands 3:45 pm Perth time; add 2 hours

B **6** $3(2n + 5) = 51$, $2n + 5 = 17$

2 Week 2 answers

1. 0.0068 first significant figure is 6; next digit 2 rounds down
2. **\$1020** 15% of 1200 = 180
3. **8 L/100 km** $\frac{24}{300} \times 100$
4. $x^2 - 2x - 24$ $x^2 - 6x + 4x - 24$
5. **212.5 km** 85×2.5
6. $t = \dfrac{v - u}{a}$ subtract u, then divide by a
7. $x = 19$ $x - 4 = 15$
8. **17** $a + (a + 5) = 39$
9. 2 $\frac{11-3}{5-1} = \frac{8}{4}$
10. $V = 900 - 150n$; **after 6 years** $150n = 900$
11. **24 cm** $x^2 = 625 - 49 = 576$
12. $x = 8$ **cm** $x = 16 \times 0.5$
13. 36π **cm²** $r = 6$: $\pi \times 36$
14. 200π **cm³** $\pi \times 25 \times 8$
15. ± 0.5 **cm** half the precision of 1 cm
16. **\$1456** $38 \times 32 + 5 \times 48 = 1216 + 240$
17. **\$5618** $5000 \times 1.06 = 5300$; 5300×1.06
18. 15 9, 11, 12, **14, 16**, 17, 19, 21
19. $\frac{1}{12}$ (4, 6), (5, 5), (6, 4): $\frac{3}{36}$
20. **A–C–B–D–E, 10 min** $2 + 1 + 5 + 2$; A–B–D–E takes 11

B **12 cm by 5 cm** $l + w = 17$ and $l^2 + w^2 = 169$

Answers

3 Week 3 answers

1. 3.2×10^{-5} move the decimal point 5 places right
2. 12.5% $\frac{80\,000}{640\,000} = \frac{1}{8}$
3. **$200, $300, $400** 9 parts: $100 each
4. $3x(2x - 3)$ HCF = $3x$
5. $S = 440$ 10×44
6. $r = \sqrt{\frac{A}{\pi}}$ divide by π, then take the square root
7. $x = 8.5$ $6x + 3 = 4x + 20$, $2x = 17$
8. **20** $25a + 15(40 - a) = 800$, $10a = 200$
9. $y = 3x - 5$ $1 = 3(2) + c$
10. **40** $9n = 200 + 4n$, $5n = 200$
11. **3.7 m** $\sqrt{1.44 + 12.25} = \sqrt{13.69}$
12. **≈ 31 m** $h = 50 \times 0.62$
13. **80π cm²** $2\pi(16) + 2\pi(4)(6) = 32\pi + 48\pi$
14. **24π cm³** $\frac{1}{3} \times \pi \times 9 \times 8$
15. 1% $\frac{0.5}{50} \times 100\%$
16. **$810** $450 + 360$
17. **$13 500** $24\,000 \times 0.75 = 18\,000$; $\times 0.75$
18. **11.5** $Q_1 = 7.5$, $Q_3 = 19$
19. $\frac{1}{3}$ $\frac{6}{10} \times \frac{5}{9} = \frac{30}{90}$
20. **9 h 40 min** $145 \div 15 = 9\frac{2}{3}$ h

B $n = 8$ $64 + 81 = 145$

4 Week 4 answers

1. 4×10^4 $4.8 \div 1.2 = 4$; 10^{6-2}
2. **$600** $660 \div 1.1$
3. **25 m/s** $90\,000 \div 3600$
4. $4x^2 - 12x + 9$ $(2x)^2 - 2(2x)(3) + 3^2$
5. $A = 2205$ 2000×1.1025
6. $h = \frac{3V}{\pi r^2}$ multiply by 3, then divide by πr^2
7. $x = 4$ $5 - 2x = 3x - 15$, $20 = 5x$
8. **31 and 19** $2x = 62$
9. **Yes** $2 \times (-\frac{1}{2}) = -1$
10. $h = 30 - 2.5t$**; after 8 hours** $2.5t = 20$
11. **130 cm** $\sqrt{14\,400 + 2500} = \sqrt{16\,900}$
12. $c = 7$ $c^2 = 9 + 64 - 2(3)(8)(0.5) = 49$
13. **6π cm²** $\frac{60}{360} \times \pi \times 36$
14. **36π cm³** $\frac{4}{3} \times \pi \times 27$
15. **7.35 cm and 7.45 cm** absolute error = 0.05 cm
16. **$65 500** $68\,000 - 2500$
17. **10% simple ($2000 vs $1881)** $10\,000 \times 1.09^2 = 11\,881$
18. **the median (5.5)** the outlier 40 pulls the mean up to about 11
19. **0.3** 0.6×0.5
20. **13 km** choose CD 2, AB 3, BC 4, DE 4

B $x = 5$ $x^2 + 3x - 40 = 0$, $(x + 8)(x - 5) = 0$; $x = -8$ gives a negative width

Answers

5 Week 5 answers

1. **26 800** the 4th figure is 4, so round down
2. **$72 100** $70\,000 \times 1.03$
3. **the 2 L drink** $2.00 per L vs $1.80 per L
4. $2x - 25$ $x^2 - 25 - x^2 + 2x$
5. $C = 10$ $1.8C = 18$
6. $b = \dfrac{P}{2} - a$ divide by 2, then subtract a
7. $x = 3$ $6x - 2 - x - 4 = 9$, $5x = 15$
8. **140 m²** width 10, length 14
9. $(4,\ 0)$ **and** $(0,\ 3)$ put $y = 0$, then $x = 0$
10. **260 USD; 200 AUD** 400×0.65; $130 \div 0.65$
11. **10 cm; 26 cm** $\sqrt{36 + 64} = 10$; $\sqrt{100 + 576} = 26$
12. $\cos\theta = \frac{12}{13}$, $\tan\theta = \frac{5}{12}$ opposite $= \sqrt{169 - 144} = 5$
13. **88 m²** $12 \times 9 - 5 \times 4 = 108 - 20$
14. **300 kL** $25 \times 10 \times 1.2 = 300\ \text{m}^3$
15. 0.25% absolute error 5 g; $\frac{5}{2000} \times 100\%$
16. **$1050** 17.5% of $6000
17. **$4200** $9000 - 4 \times 1200$
18. **strong, positive, linear** direction, strength, form
19. 0.46 $\frac{92}{200}$
20. **$322** $258 + 160 + 120 = 538$

B **4 years** values: 16 000, 12 800, 10 240, 8192

6 Week 6 answers

1. 1.2×10^3 12×10^2
2. **$120** $150 \div 1.25$
3. **2 h 40 min** $2400 \div 15 = 160$ min
4. $(x - 5)(x + 4)$ $-5 \times 4 = -20$, $-5 + 4 = -1$
5. $a = 8$ $60 = 3(a + 12)$
6. $r = \sqrt[3]{\dfrac{3V}{4\pi}}$ multiply by 3, divide by 4π, then cube root
7. $x = 3, y = 2$ $4x + 6x - 12 = 18$
8. **5** $2(12 - n) + 5n = 39$, $3n = 15$
9. $y = -\frac{1}{2}x + \frac{9}{2}$ gradient $-\frac{1}{2}$: $4 = -\frac{1}{2} + c$
10. **$459** $99 + 360$
11. **13 cm** $\sqrt{25 + 144}$
12. $b \approx 15.4$ $b = \frac{10 \times 0.77}{0.5}$
13. **360 cm²** $100 + 4 \times \frac{1}{2} \times 10 \times 13$
14. 144π **cm³** $\frac{2}{3} \times \pi \times 216$
15. **9700.25 m²** 120.5×80.5
16. **$9000** $0.12 \times 75\,000$
17. **$1261** $8400 \to 8820 \to 9261$
18. 13 $(120 - 29) \div 7 = 91 \div 7$
19. $\frac{125}{216}$ $\left(\frac{5}{6}\right)^3$
20. **4:00 am Friday** difference 17 hours: $9\ \text{pm} - 17\ \text{h}$

B **$310 more** 13 310 vs 13 000